Relative Risk from Two-Way Tables

A **two-way table** sorts data by two categories at once — one category down the rows, one across the columns. Each row and column has a **total**, and the bottom-right cell is the **grand total** (everyone in the data set).

1 Two-Way Tables: The Basics

Complete each table, then answer the questions.

1. Favourite subject

220 students were asked their favourite subject out of Maths, Art and PE.

	Maths	Art	PE	**Total**
Year 9	45	30	25	
Year 10	35	40	45	
Total				

1. How many Year 10 students chose Art? _____
2. What fraction of all 220 students are in Year 9? _____
3. What percentage of Year 9 students chose PE? _____
4. Of all the students who chose Maths, what fraction are in Year 10? _____

2. Pet ownership

Households in an urban and a rural area were asked what pets they own.

	Dog	Cat	No pet	**Total**
Urban	20	35	45	
Rural	50	15	10	
Total				

1. How many rural households own a cat? _____
2. What fraction of urban households have no pet? _____
3. What percentage of all households own a dog? _____
4. Of all the households that own a cat, what fraction are rural? _____

2 Relative Risk

In a two-way table about risk, the rows are usually two **groups** (e.g. exposed/not exposed, treated/untreated) and the columns are whether an **outcome** happened or not.

$$\text{risk (for a group)} = \frac{\text{number in that group } \textit{with} \text{ the outcome}}{\text{total number in that group}}, \qquad \text{RR} = \frac{\text{risk in Group A}}{\text{risk in Group B}}.$$

Note: risk uses the **row** total for that group, *not* the grand total. If $RR = 1$ the risk is the same in both groups; if $RR > 1$ Group A has the higher risk; if $RR < 1$ Group A has the lower risk.

For each table below: complete the table, then find the risk for each group, the relative risk, and interpret it.

1. Vaccination and flu

1000 people took part in a study. 500 received a flu vaccine and 500 did not. The table shows how many of each group caught the flu that winter.

	Flu	No flu	**Total**
Vaccinated	40		500
Unvaccinated	100		500
Total			

1. Risk of flu for the vaccinated group. _____
2. Risk of flu for the unvaccinated group. _____
3. Relative risk of flu for the vaccinated group compared to the unvaccinated group. _____
4. Interpret your answer to part (c) in a sentence.

_____

2. Energy drinks and sleep problems

A study of 500 teenagers looked at whether regularly drinking energy drinks is linked to sleep problems.

	Sleep problems	No sleep problems	**Total**
Regular energy drink user	72		200
Not a regular user	36		300
Total			

1. Risk of sleep problems for regular energy drink users. _____
2. Risk of sleep problems for people who are not regular users. _____
3. Relative risk of sleep problems for regular users compared to non-users. _____
4. Interpret your answer to part (c) in a sentence.

_____

3. Seatbelts and injury

Police records of 1000 car accidents were checked to see whether wearing a seatbelt made a difference to whether a person was injured.

	Injured	Not injured	**Total**
Wearing a seatbelt	30		600
Not wearing a seatbelt	80		400
Total			

1. Risk of injury for people wearing a seatbelt. _____
2. Risk of injury for people not wearing a seatbelt. _____
3. Relative risk of injury for seatbelt wearers compared to non-wearers. _____
4. Interpret your answer to part (c) in a sentence.

_____

4. Exercise and heart disease

740 adults over 50 were tracked over several years to see whether regular exercise was linked to heart disease.

	Heart disease	No heart disease	**Total**
Regular exercise	25		500
No regular exercise	60		240
Total			

1. Risk of heart disease for people who exercise regularly. _____
2. Risk of heart disease for people who do not exercise regularly. _____
3. Relative risk of heart disease for the regular-exercise group compared to the no-exercise group. _____
4. Interpret your answer to part (c) in a sentence.

_____

5. Sunscreen and sunburn

400 beachgoers were surveyed about sunscreen use and whether they got sunburnt that day.

	Sunburn	No sunburn	**Total**
Used sunscreen	15		200
Did not use sunscreen	30		200
Total			

1. Risk of sunburn for people who used sunscreen. _____
2. Risk of sunburn for people who did not use sunscreen. _____
3. Relative risk of sunburn for sunscreen users compared to non-users. _____
4. Interpret your answer to part (c) in a sentence.

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Reflection: Which of the five tables above show a *protective* factor (one that lowers risk), and which show a factor that *increases* risk?

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